

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example:

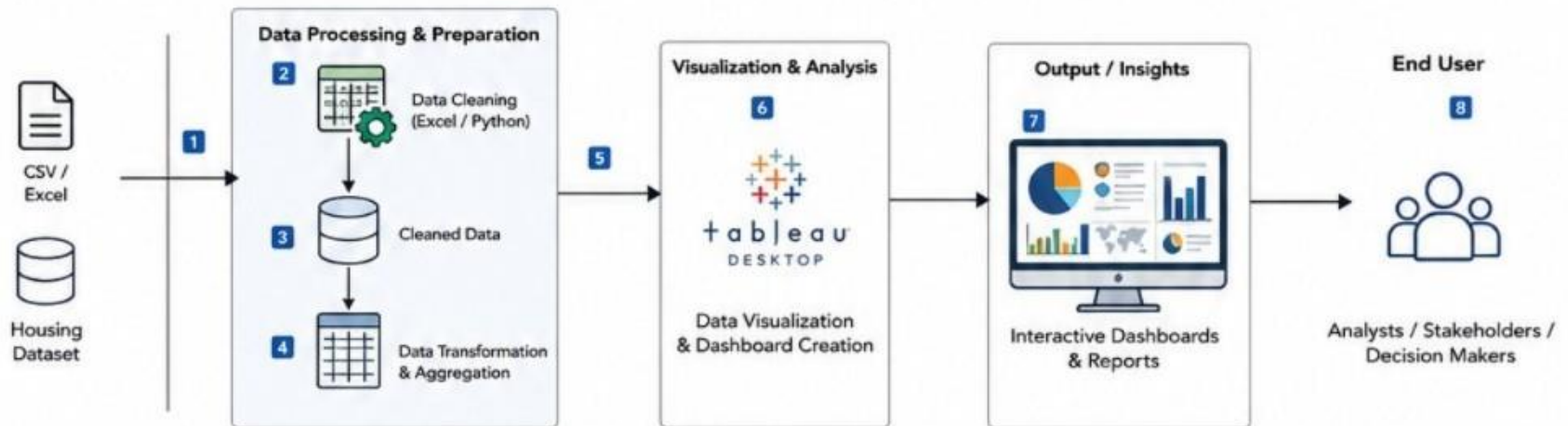


Table 1 : components and technologies

S.No	Component	Description	Technology
1	Front-End Dashboard	Interactive dashboards and visualizations for housing market analysis	
2.	Application Logic-1	Data cleaning, preprocessing, and transformation of housing datasets	Java / Python
3.	Application Logic-2	Statistical analysis and feature correlation analysis	IBM Watson STT service
4.	Application Logic-3	Data integration and dashboard interaction logic	IBM Watson Assistant
5.	Database	Storage of housing datasets and processed data	MySQL, NoSQL, etc.
6.	Cloud Database	Cloud storage and data sharing services	IBM DB2, IBM Cloudant etc.
7.	File Storage	Storage of datasets, reports, dashboards, and project files	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Housing and real estate data collection (if required)	IBM Weather API, etc.
9.	External API-2	Geographical and location-based data visualisation	Aadhar API, etc.
10.	Machine Learning Model	Predictive analysis of housing prices and market trends (optional)	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Deployment and publishing of dashboards and reports	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source libraries used for data preprocessing, analysis, and visualization support.	Python, Pandas, NumPy, Matplotlib, Seaborn
2.	Security Implementations	Secure handling of datasets and controlled access to dashboards and reports.	Tableau User Authentication, Role-Based Access Control (RBAC), HTTPS Encryption
3.	Scalable Architecture	The system can accommodate larger housing datasets and additional visualizations without major architectural changes.	Tableau Cloud, Modular Dashboard Design, Python-based Data Processing
4.	Availability	Dashboards and reports are accessible online and can be shared with multiple users at any time.	Tableau Public/Tableau Cloud, Cloud Storage Services
5	Performance	Efficient processing and visualization of housing data with optimized queries and interactive filters for faster response times.	Tableau Extracts (.hyper), Data Caching, Python Data Optimization Techniques